

Jibesh Patra

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Education

Ph.D. in Computer Science

2015–2021

Thesis: Analyzing Code Corpora to Improve the Correctness and Reliability of Programs 📄

Part of the work was done at TU Darmstadt, Germany.

University of Stuttgart

Stuttgart, Germany

Master's in Computer Science

2011–2013

National Institute of Technology

(NIT)

Durgapur, India

Bachelor's in Computer Science

2005–2009

West Bengal University of

Technology

Bankura, India

Positions and Experience



Indian Institute of Technology Kharagpur (IIT KGP)

Kharagpur, India

ASSISTANT PROFESSOR

May 2024 - Present

Primary responsibilities include teaching graduate and undergraduate students. Research topics on the applications of Machine Learning in Software Engineering, Programming Languages.



Honda Research Institute

Offenbach am Main, Germany

SENIOR SCIENTIST

Feb. 2023 - May 2024

Research topics on the applications of Machine Learning in Software Engineering. Improve the overall quality of code and integrate better Software Engineering practices.

SAP SE

Walldorf, Germany

SENIOR DEVELOPER

May 2021 - Jan. 2023

I worked on improving the code quality of a large JavaScript code base by statically analyzing and refactoring commonly misused patterns. My work has reduced the overall build time of a large build infrastructure by 45%.



University of Stuttgart

Stuttgart, Germany

RESEARCH ASSISTANT

Mar. 2019 - Apr. 2021

Moved along with the Software Lab research group. I worked on testing and program analysis, with focus on checking the correctness of programs. The highlights of my work are:

- Found bugs in real-world Python code using Deep Learning. The first step was to instrument a corpus of Python files and to execute them. This dynamic analysis step resulted in the training data for the Deep Learning model.
- Introduced realistic bugs in JavaScript programs that boosts the performance of learning based bug-finding approaches.
- I worked on predicting type signatures of JavaScript functions using Deep Learning. This work leveraged the natural language information present in variable names to successfully predict the return types.



TU Darmstadt

Darmstadt, Germany

RESEARCH ASSISTANT

Mar. 2015 - Feb. 2019

As part of the Software Lab research group, I worked on testing and program analysis, with focus on JavaScript. The highlights of my work are:

- In many cases, JavaScript libraries included in web pages write to the same global accesspath which may lead to unexpected behaviors for users. I investigated a way to detect such cases and found that many popular JavaScript libraries actually conflict when included together.
- I generated large number of JavaScript programs to fuzz test JavaScript engines like V8 and SpiderMonkey and found bugs in them.

RESEARCH INTERN

Jun. 2018–Sep. 2018

The highlights of my work are:

- I worked on the applications of Machine Learning methods with source code as input.
- I used Recurrent Neural Networks (RNNs) to summarize code.

Bangalore, India

RESEARCH INTERN

Jun. 2017–Sep. 2017

The highlights of my work are:

- I understood simple protocols for secure multi-party computation and the syntax of a new programming language that implements such protocols.
- I specified the semantics of the new language.

**Max Planck Institute for Software Systems**

Kaiserslautern, Germany

INTERN

Oct. 2013–Dec. 2013

I explored program slicing and topics like causation and counterfactuals, to reason about their applications in ranking of program slices for asynchronous programs.

**Max Planck Institute for Heart and Lung Research**

Bad Nauheim, Germany

INTERN

Jun. 2013–Sep. 2013

I worked on the next generation sequencing area of Bioinformatics.

- I performed quality control of DNA sequence files.
- I set up a web server for the analysis process.
- I implemented an interface between the web server and an external cluster where the jobs can be submitted, making the analysis process faster.

Remote

INTERN

May. 2012–Aug. 2012

I contributed to OpenMRS, an open source medical record system and integrated a file upload/download feature in one of its modules.

**High School**

West Bengal, India

ASSISTANT TEACHER

Aug. 2010–Aug. 2011

Apr. 2014–Jan 2015

I introduced coding to high school students.

**Wipro Technologies**

Hyderabad, India

PROJECT ENGINEER

May. 2010–Aug. 2010

As part of the training program, I learned system programming in Linux and implemented a project involving it.

Research Papers

TypePatrol: Adversarial Testing for Uncovering Security-Relevant Type Inconsistencies in JavaScript Libraries  *International Symposium on Empirical Software Engineering and Measurement (ESEM)*

[Core Rank A]

SAI MALLIKARJUN GALIPELLI, CRISTIAN-ALEXANDRU STAIU, JIBESH PATRA

2026

We find a large number of type inconsistencies in TypeScript and JavaScript libraries through dynamic analysis.

Nalin: Learning from Runtime Behavior to Find Name-Value Inconsistencies in Jupyter Notebooks 

International Conference on Software Engineering (ICSE)

[Core Rank A*]

JIBESH PATRA, MICHAEL PRADEL

2022

We use data obtained from dynamic program analysis to train a neural model that finds bugs in real world Python code.

Semantic Bug Seeding: A Learning-Based Approach for Creating Realistic Bugs

JIBESH PATRA, MICHAEL PRADEL

Distinguished Paper Award



We generate large bug datasets by imitating real world bugs. We find that the generated bugs are useful in training learning-based bug detectors.

2021

A Survey of Compiler Testing

ACM Computing Surveys (CSUR)
[Impact Factor 10.2]

JUNJIE CHEN, JIBESH PATRA, MICHAEL PRADEL, YINGFEI XIONG, HONGYU ZHANG, DAN HAO, LU ZHANG

Our work provides a comprehensive survey of various techniques used to test compilers.

2020

NL2Type: Inferring JavaScript Function Types from Natural Language Information

International Conference on
Software Engineering (ICSE)
[Core Rank A*]

RABEE SOHAIL MALIK, JIBESH PATRA, MICHAEL PRADEL

We train neural models that aid developers by suggesting types in not-yet annotated JavaScript code. This work has directly inspired an approach called 'TypeWriter' that has been deployed at Facebook.

2019

ConflictJS: Finding and Understanding Conflicts Between JavaScript Libraries

International Conference on
Software Engineering (ICSE)
[Core Rank A*]

JIBESH PATRA, POOJA N. DIXIT, MICHAEL PRADEL

Using dynamic program analysis, our work finds that when included together, even popular JavaScript libraries can cause unexpected behavior for users.

2018

Automatically Reducing Tree-Structured Test Inputs

International Conference on
Automated Software Engineering
(ASE)
[Core Rank A*]

SATIA HERFERT, JIBESH PATRA, MICHAEL PRADEL

Our work leverages large corpora of code to reduce test inputs.

2017

Learning to Fuzz: Application-Independent Fuzz Testing with Probabilistic, Generative Models of Input Data

Technical Report, TU Darmstadt

JIBESH PATRA, MICHAEL PRADEL

Our work learns probabilistic models of code from a large corpus and generates data for fuzz testing.

2016

Congestion Balancing Global Router

VLSI Design and Test (VDAT)

SHYAMAPADA MUKHERJEE, JIBESH PATRA, SUCHISMITA ROY

We propose a solution to global routing for VLSI circuits.

2013

Skills

Statically Typed Languages	C, Java
Dynamically Typed Languages	JavaScript, Python
Others	PyTorch, NumPy, pandas, Node.js, $\text{\LaTeX}$, HTML
Natural Languages	English (fluent), German (intermediate), Bangla (fluent), Odia (fluent)

Teaching, Mentoring and Service

Instructor/Co-Instructor

IIT Kharagpur, India

SOFTWARE ENGINEERING
SOFTWARE ENGINEERING LABORATORY
OBJECT-ORIENTED SYSTEMS
THEORY & APPLICATIONS OF BLOCKCHAIN
PROGRAMMING & DATA STRUCTURES LAB

2024-Present

Teaching Assistant

TU Darmstadt, Germany
University of Stuttgart, Germany

PROGRAM TESTING AND ANALYSIS
PROGRAMMING PARADIGMS
ANALYZING SOFTWARE USING DEEP LEARNING
MACHINE LEARNING FOR PROGRAMMING

2015-2021

Program/Evaluation Committee

CAIN'25 - Industry Track - Program Committee
FSE'25 - Poster Track - Program Committee
ISSTA'17 - Artifact Evaluation Committee

Student Volunteer

ECOOP
ESEC/FSE

2016

2017

Served as a student volunteer for conferences such as European Conference on Object-Oriented Programming (ECOOP) and Foundations of Software Engineering (ESEC/FSE).

Assistant Teacher

Govt. of West Bengal, India

2010-2011

2014-2015

Introduced coding to high school students.

Awards and Grants

Distinguished Paper Award

ESEC/FSE

SEMANTIC BUG SEEDING: A LEARNING-BASED APPROACH FOR CREATING REALISTIC BUGS 

2021

Received the award for my paper at the European Software Engineering Conference and Symposium on the Foundations of Software Engineering conference.

ACM Travel Grant

ICSE

CONFLICTJS: FINDING AND UNDERSTANDING CONFLICTS BETWEEN JAVASCRIPT LIBRARIES

2018

Received a travel grant of 620\$ to present my paper at the International Conference on Software Engineering, held in Gothenburg, Sweden.

Prize for Good Supervision

TU Darmstadt

SUPERVISION FOR PROGRAM TESTING AND ANALYSIS COURSE 

2017

Received the award *Preis für gute Betreuung*, along with my fellow teaching assistants, based on feedback from the participating students.

Distinguished Poster Award

ECOOP

LANGUAGE-INDEPENDENT FUZZ TESTING WITH PROBABILISTIC, GENERATIVE MODELS 

2016

Received the award for my poster at the European Conference on Object-Oriented Programming.